ECE 590
Neural-Network-Based Large Language Models

Tuesdays and Thursdays 1:25-2:40
Fitzpatrick Schiciano A 1464

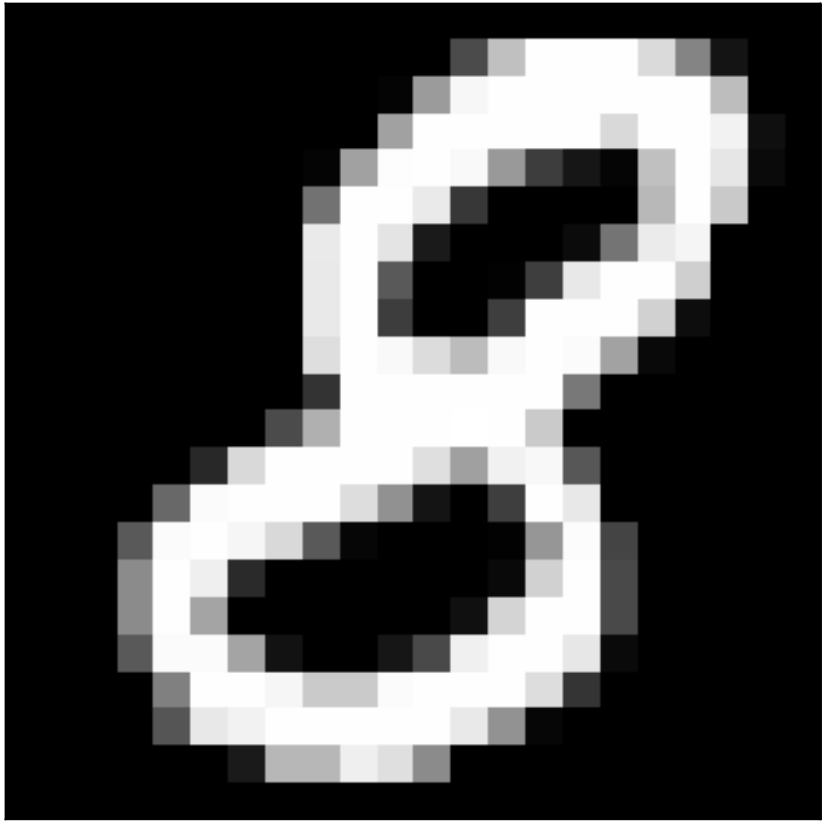
ECE 590
Neural-Network-Based Large Language Models

Lecture 2
01/13/2026

The MNIST Dataset

- The Modified National Institute of Standards and Technology (MNIST) contains pictures of handwritten digits (0,1,2,…)
- Want to be able to tell what digit each image is (*e.g.*, optical character recognition)

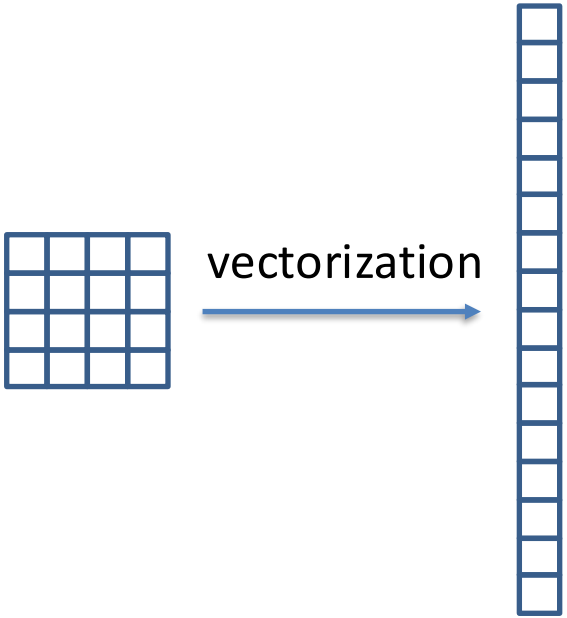
Images are Encoded as Numbers



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	118	200	223	155	155	23	0	0	0	0	0
0	0	0	0	0	0	0	0	6	124	214	253	253	253	253	254	229	213	67	0	0	0
0	0	0	0	0	0	0	43	198	253	254	253	247	175	175	236	253	253	108	0	0	0
0	0	0	0	0	0	9	212	253	253	224	58	19	0	0	16	139	247	247	71	0	0
0	0	0	0	0	0	58	254	254	186	14	0	0	0	0	0	0	139	254	118	0	0
0	0	0	0	0	0	58	253	183	8	0	0	0	0	0	0	36	222	253	118	0	0
0	0	0	0	0	0	54	250	128	2	0	0	0	6	71	192	237	253	247	71	0	0
0	0	0	0	0	0	0	213	253	50	0	18	123	198	253	254	253	247	85	0	0	0
0	0	0	0	0	0	0	69	241	227	136	200	253	253	253	254	192	34	0	0	0	0
0	0	0	0	0	0	0	0	178	254	256	254	254	254	149	59	0	0	0	0	0	0
0	0	0	0	0	0	0	101	253	253	254	253	253	253	42	0	0	0	0	0	0	0
0	0	0	0	0	8	138	247	253	243	159	196	243	253	199	0	0	0	0	0	0	0
0	0	0	0	12	183	253	253	253	50	0	0	49	253	214	0	0	0	0	0	0	0
0	0	0	0	180	254	253	215	50	2	0	0	71	253	214	0	0	0	0	0	0	0
0	0	0	0	234	217	97	10	0	0	0	23	207	254	215	0	0	0	0	0	0	0
0	0	0	174	253	156	0	0	0	0	45	215	253	253	95	0	0	0	0	0	0	0
0	0	9	210	253	163	5	19	49	130	244	253	251	137	4	0	0	0	0	0	0	0
0	0	13	229	253	254	192	253	253	253	254	253	137	0	0	0	0	0	0	0	0	0
0	0	0	160	253	254	253	253	253	253	239	132	4	0	0	0	0	0	0	0	0	0
0	0	0	18	112	194	254	254	254	163	59	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

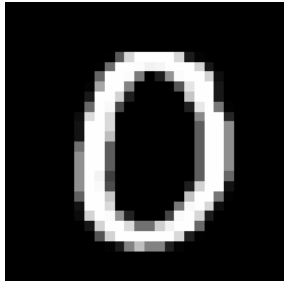
Vectorization

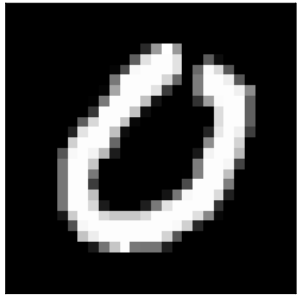
- We will start talking about deep learning *without* using the structure of the image
- We will return to this in a future lecture
- To convert an image into an unstructured set of numbers, we *vectorize it*

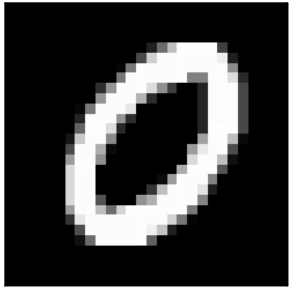


Start With The Binary Case

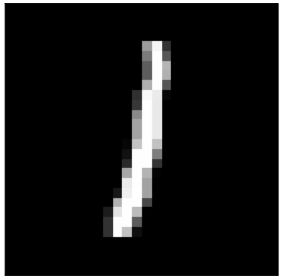
Zeros

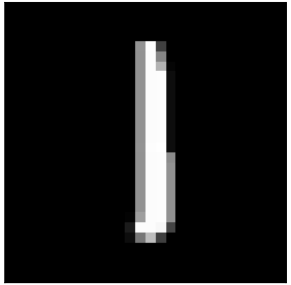


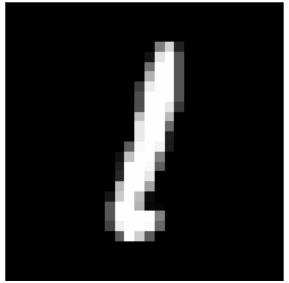




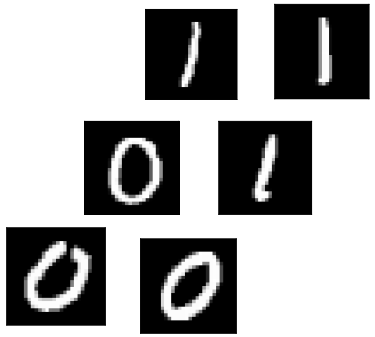
Ones



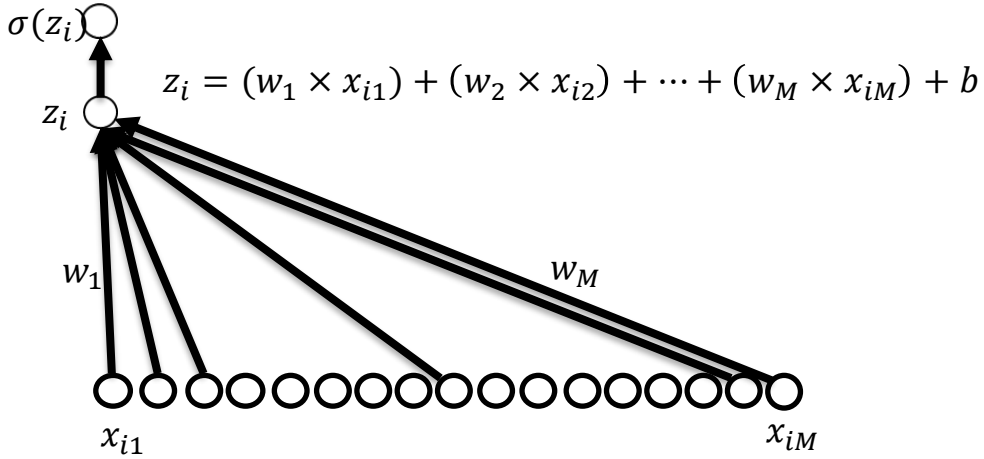




Learning on MNIST



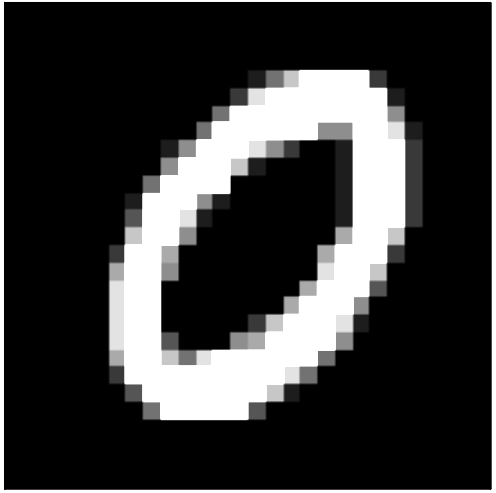
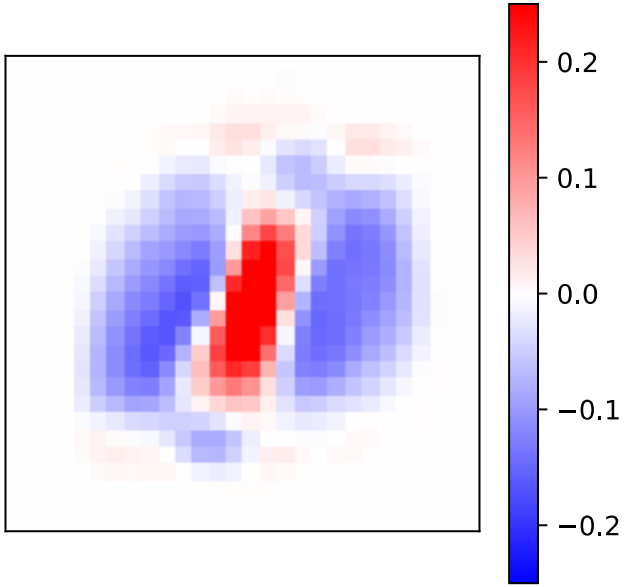
Vectorization



Logistic Regression Model (or "Network")

Learned Parameters $(b, w_1, \ldots w_M)$

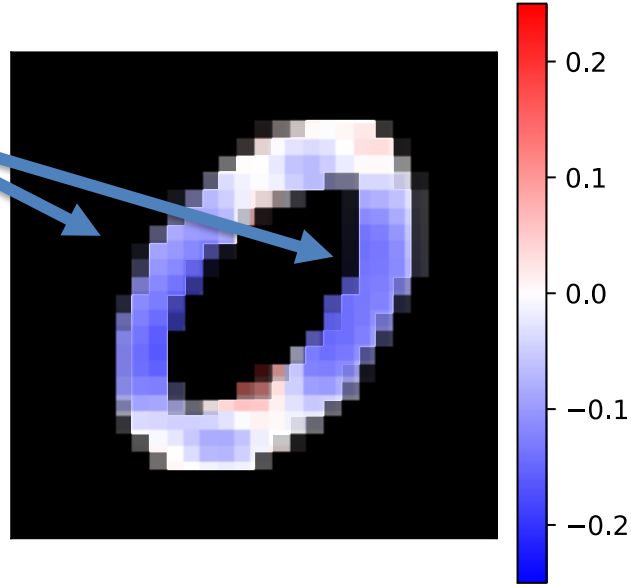
Zooming in on 0/1

$\sigma($  $\times$ 

$)$

Zooming in on 0/1

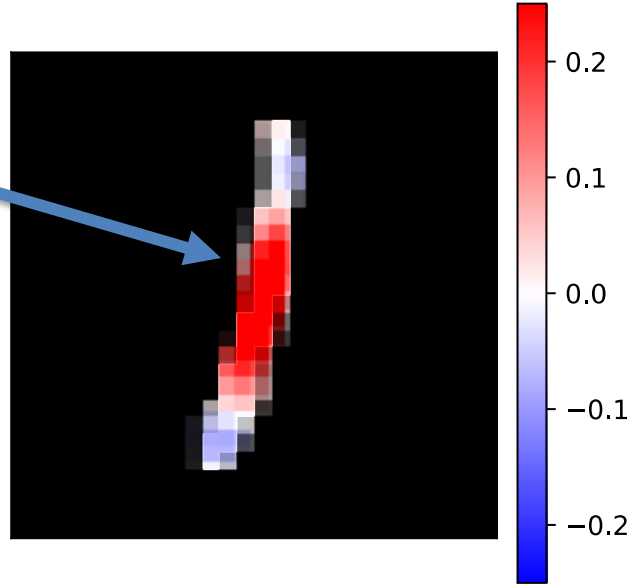
Negative Sections

$\sigma($ 

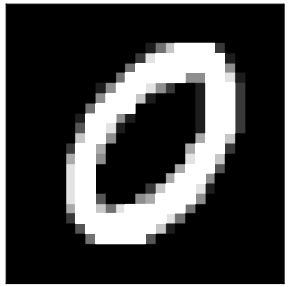
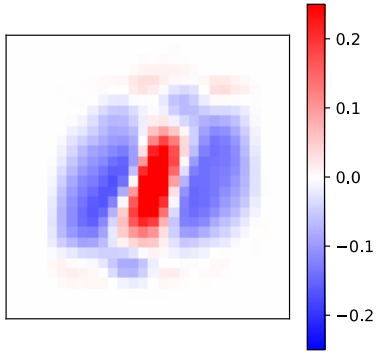
$) = 0.006$

Zooming in on 0/1

Positive Section

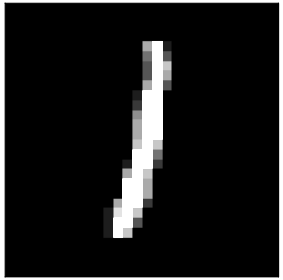
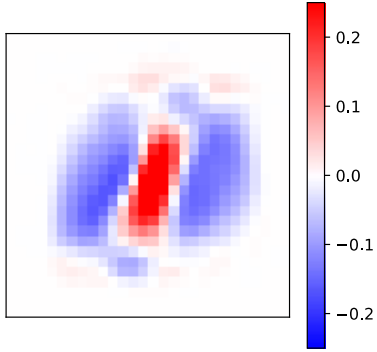
$\sigma($  $) = .991$

Learned Weights for 0/1

$\sigma($  $\times$ 

$) = .006$

We think that this is a “zero” (.6% chance it is a “one”)

$\sigma($  $\times$ 

$) = .991$

We think that this is a “one” (99.1% chance it is a “one”)

SLOWING DOWN: WHAT'S A LINEAR MODEL?

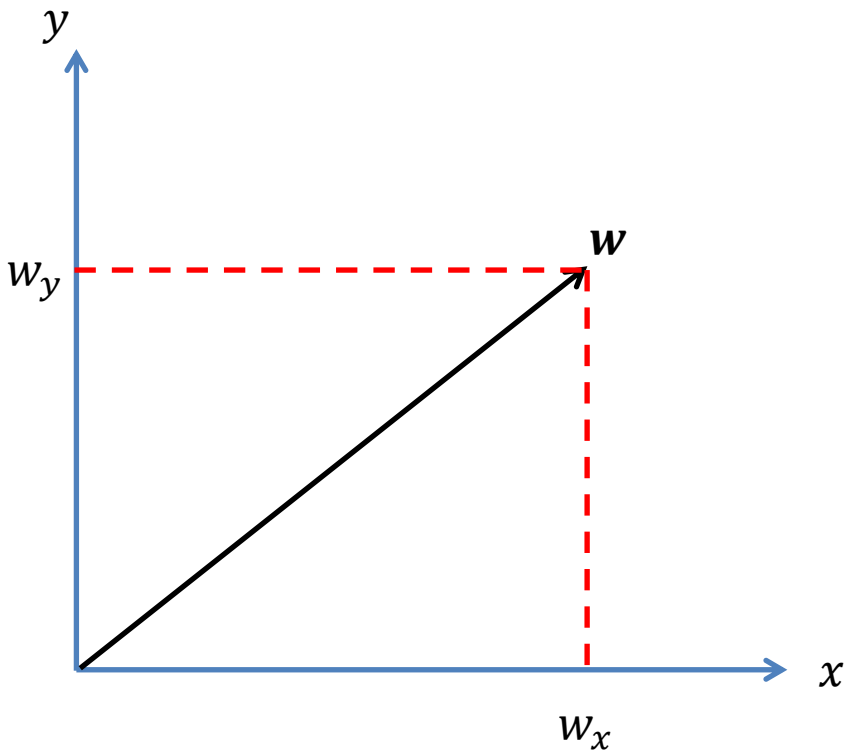
- The concept of "linear" will have multiple different notions:

  1. Project the data x onto a line, defined by vector w

  2. A linear model of the outcome along the aforementioned line

  3. Equal predicted outcomes, for points along lines perpendicular to w

- Let's walk through this stepwise

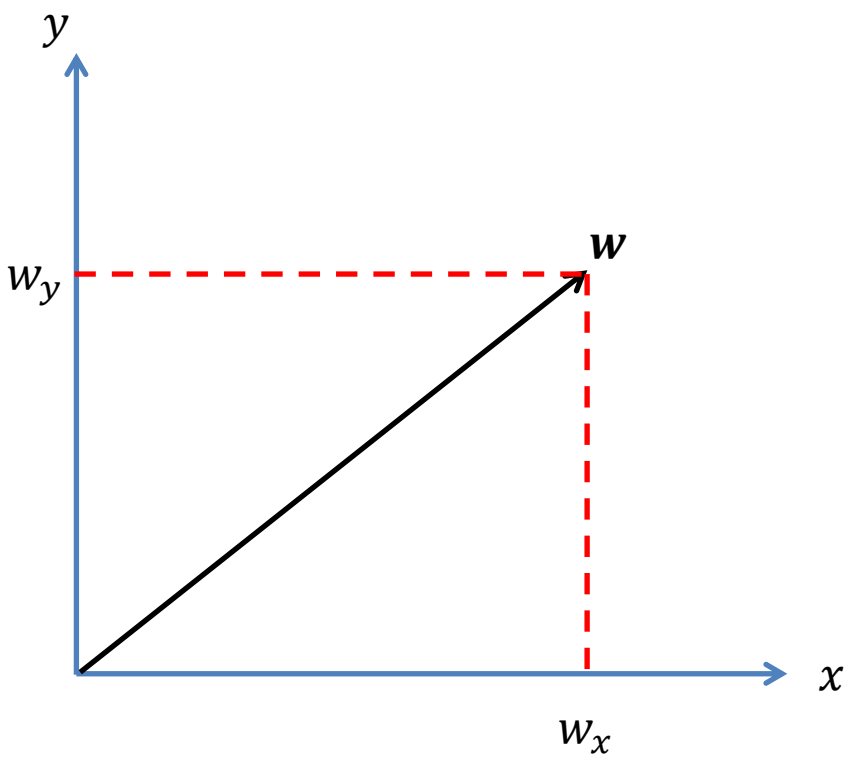


$$\boldsymbol{w} = \hat{\boldsymbol{x}} w_x + \hat{\boldsymbol{y}} w_y$$

$$\boldsymbol{w} = [w_x \; w_y] \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix}$$

$$\hat{\boldsymbol{w}} = \boldsymbol{w} / \|w\|_2$$

$$\|\boldsymbol{w}\|_2 = \sqrt{w_x^2 + w_y^2}$$



$$\boldsymbol{w} = \hat{\boldsymbol{x}} w_x + \hat{\boldsymbol{y}} w_y$$

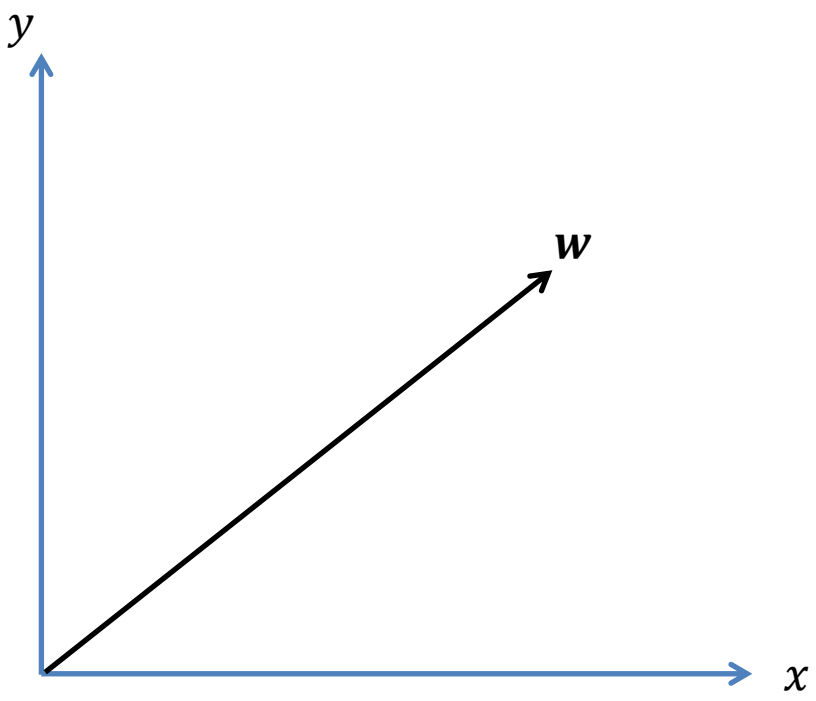
$$\boldsymbol{w} = [w_x \; w_y] \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix}$$

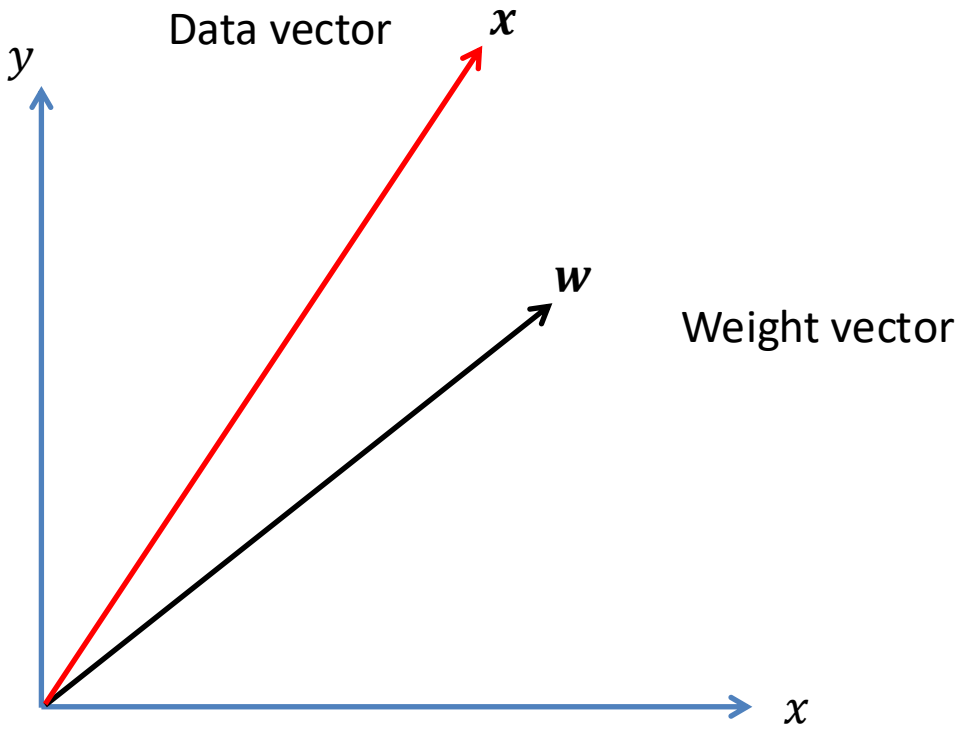
$$\hat{\boldsymbol{w}} = \boldsymbol{w} / \|w\|_2$$

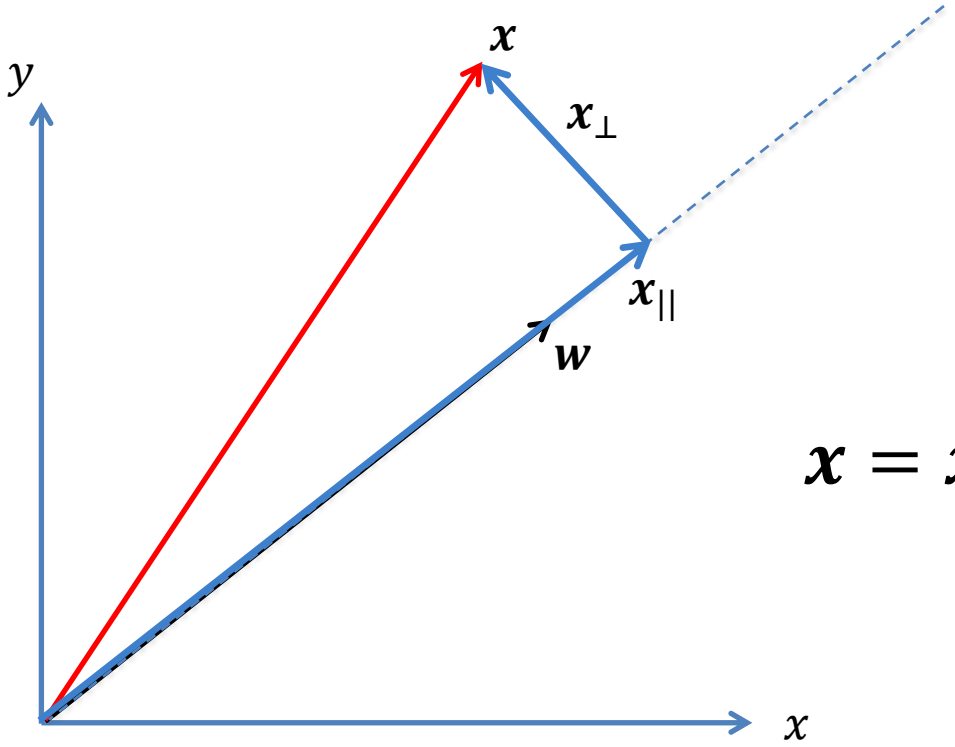
Unit vector in direction of $\boldsymbol{w}$

$$\|\boldsymbol{w}\|_2 = \sqrt{w_x^2 + w_y^2}$$

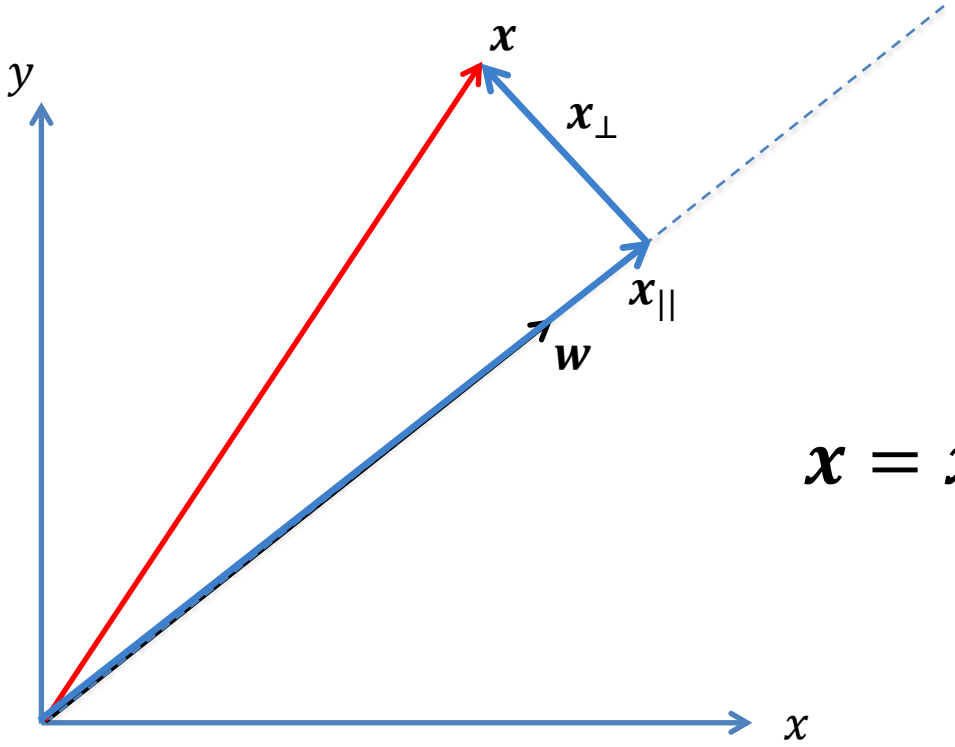
Magnitude of $\boldsymbol{w}$


y
w
x


Data vector
$\boldsymbol{x}$
y
$\boldsymbol{w}$
Weight vector
x



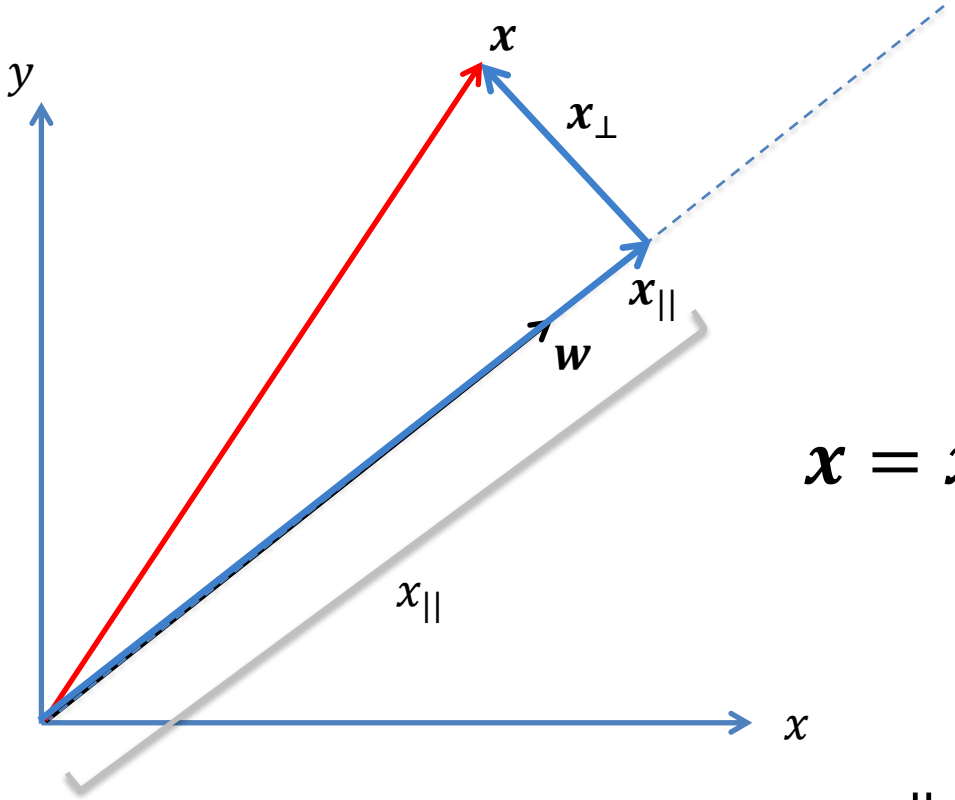
$$\boldsymbol{x} = \boldsymbol{x}_{||} + \boldsymbol{x}_\perp$$



Parallel to w

$$\boldsymbol{x} = \boldsymbol{x}_{||} + \boldsymbol{x}_\perp$$

Perpendicular to w



Parallel to w

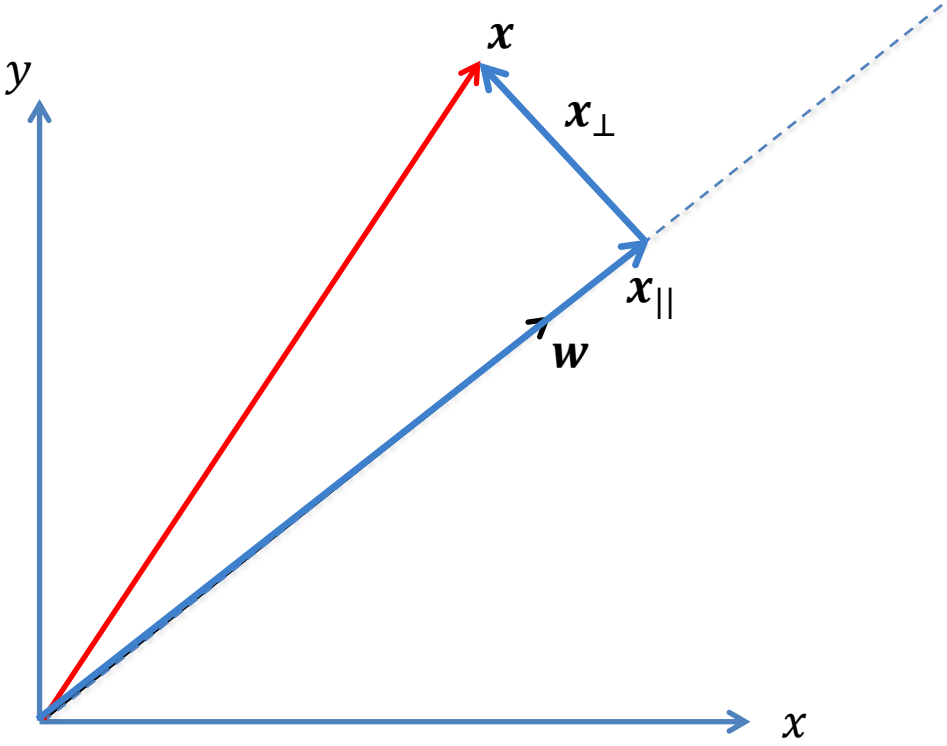
$$\boldsymbol{x} = \boldsymbol{x}_{||} + \boldsymbol{x}_{\perp}$$

Perpendicular to w

$$\hat{\boldsymbol{w}} \cdot \boldsymbol{x} = x_{||} \ , \ \text{where} \ \ x_{||} = \left\| \boldsymbol{x}_{||} \right\|_2$$

$$\hat{\boldsymbol{w}} \cdot \boldsymbol{x} = x_{||} \ , \ \text{where} \ \ x_{||} = \left\| \boldsymbol{x}_{||} \right\|_2$$

Effectively $\hat{\boldsymbol{w}}$ constituutes a function $f_{\hat{\boldsymbol{w}}}(\boldsymbol{x}) \colon \mathbb{R}^2 \to \mathbb{R}$

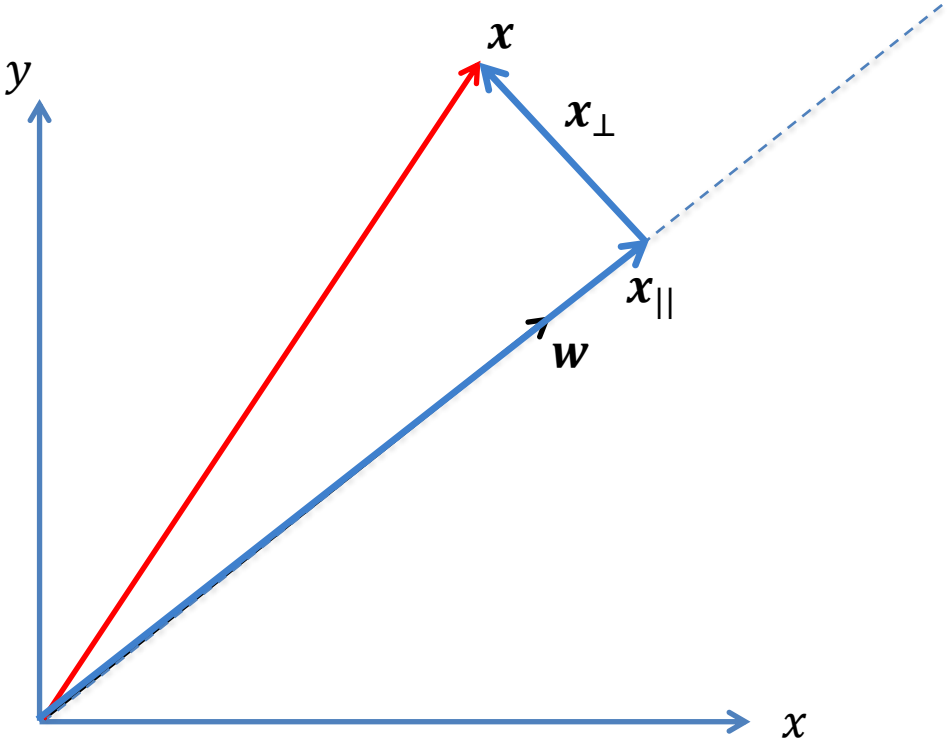


Model vector: $\boldsymbol{w} = \|\boldsymbol{w}\|_2 \hat{\boldsymbol{w}}$

Data vector: $\boldsymbol{x} = x_{||}\hat{\boldsymbol{w}} + x_\perp \hat{\boldsymbol{w}}_\perp$

$$\boldsymbol{x} = \boldsymbol{x}_{||} + \boldsymbol{x}_\perp$$

$$\hat{\boldsymbol{w}} \cdot \hat{\boldsymbol{w}}_\perp = 0$$



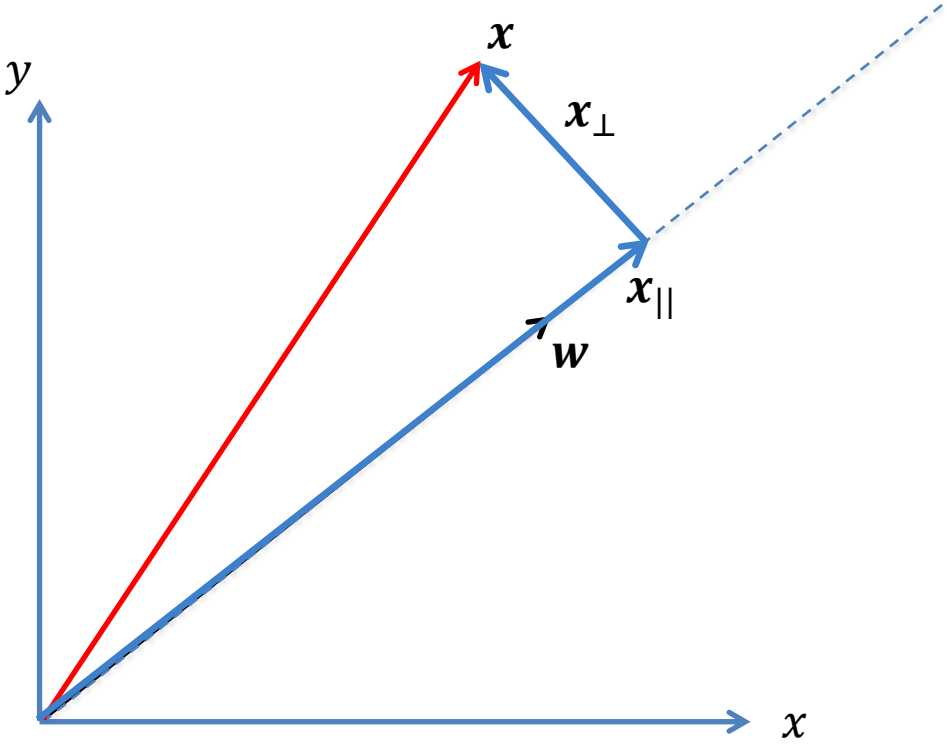
Model vector: $\boldsymbol{w} = \|\boldsymbol{w}\|_2 \hat{\boldsymbol{w}}$

Data vector: $\boldsymbol{x} = x_{||}\hat{\boldsymbol{w}} + x_\perp \hat{\boldsymbol{w}}_\perp$

$$\hat{\boldsymbol{w}} \cdot \boldsymbol{x} = x_{||}$$

$$\boldsymbol{w} \cdot \boldsymbol{x} = \|\boldsymbol{w}\|_2 x_{||}$$

$$= \|\boldsymbol{w}\|_2 \left\|x_{||}\right\|_2$$



Model vector: $\boldsymbol{w} = \|\boldsymbol{w}\|_2 \hat{\boldsymbol{w}}$

Data vector: $x = x_{||}\hat{\boldsymbol{w}} + x_{\perp}\hat{\boldsymbol{w}}_{\perp}$

Position of $\boldsymbol{x}$ on Line specified By $\boldsymbol{w}$

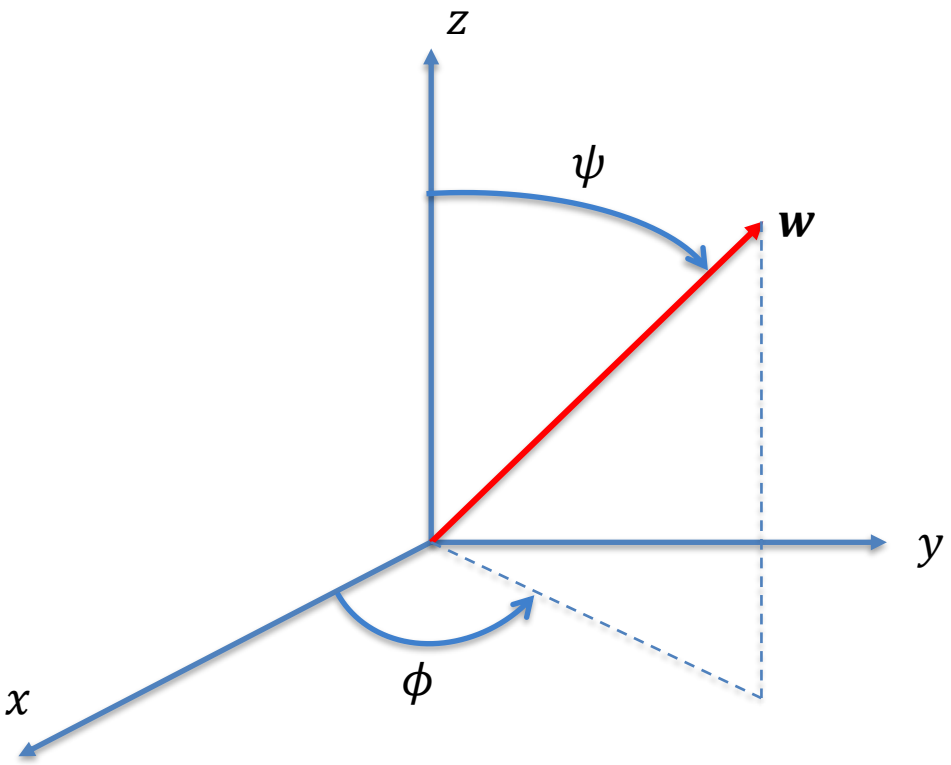
$$\boldsymbol{w} \cdot \boldsymbol{x} = \|\boldsymbol{w}\|_2 x_{||}$$

Length of $\boldsymbol{w}$

First Meaning of Linear

- Project data $\boldsymbol{x}$ onto a line define by the direction of $\boldsymbol{w}$

- Have shown line in 2D or 3D, but generalizes to lines $\boldsymbol{w}$ in N-dimensional space


z
ψ
w
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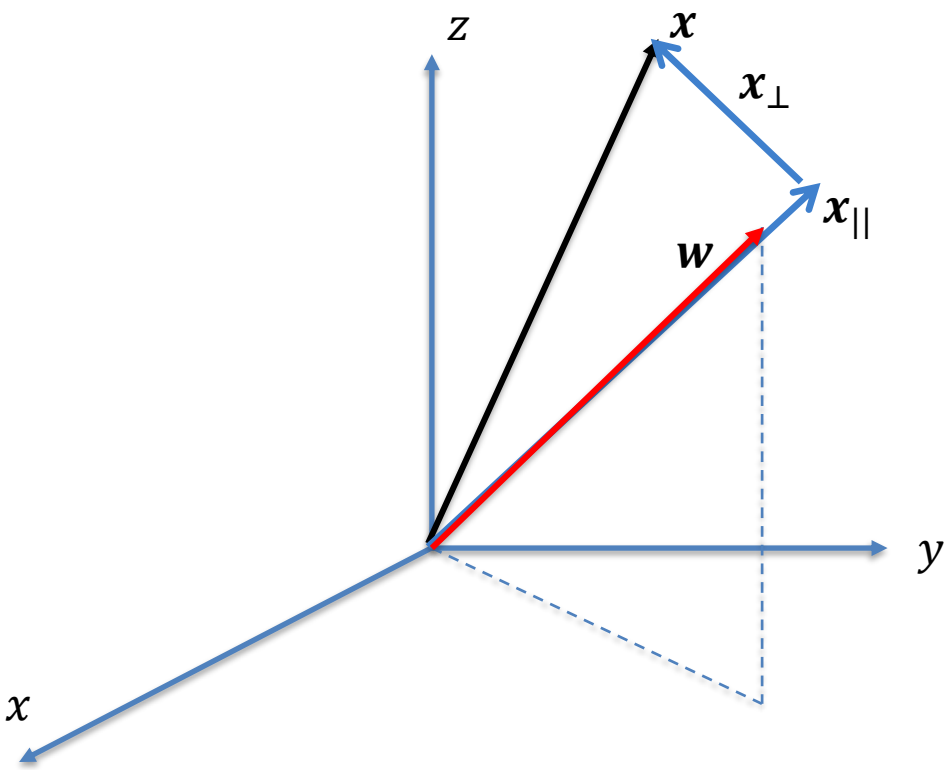
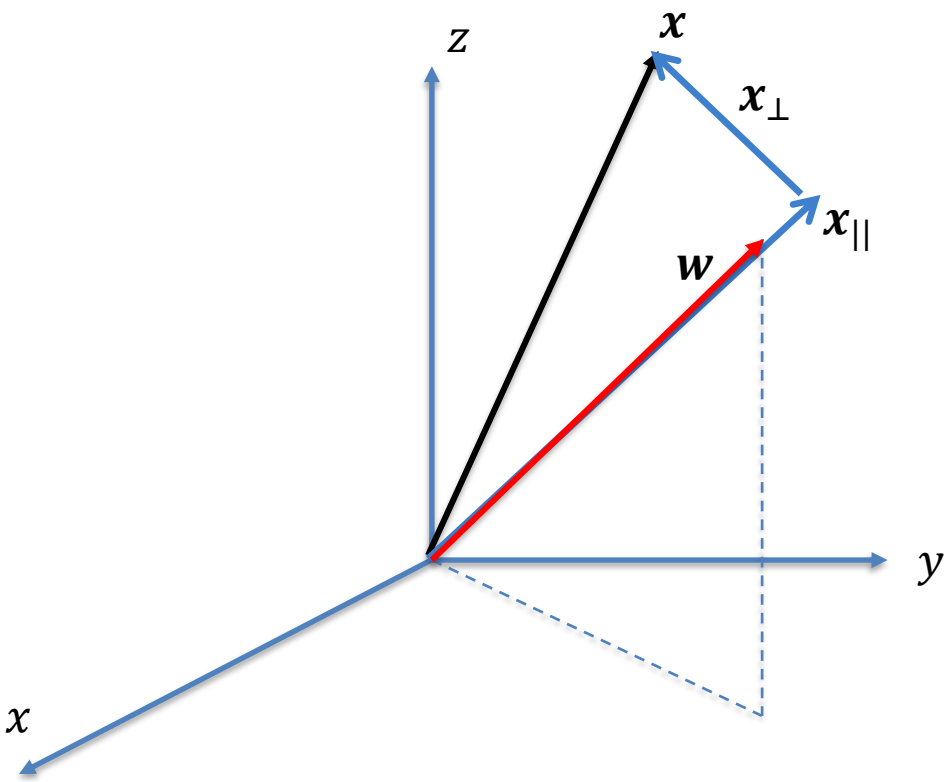
x
ϕ



$$\boldsymbol{x} = \widehat{\boldsymbol{w}} x_{||} + \widehat{\boldsymbol{w}}_\perp x_\perp$$

$$\widehat{\boldsymbol{w}} \cdot \boldsymbol{x} = x_{||}$$

Position of $\boldsymbol{x}$ after projection
Onto weight vector $\boldsymbol{w}$



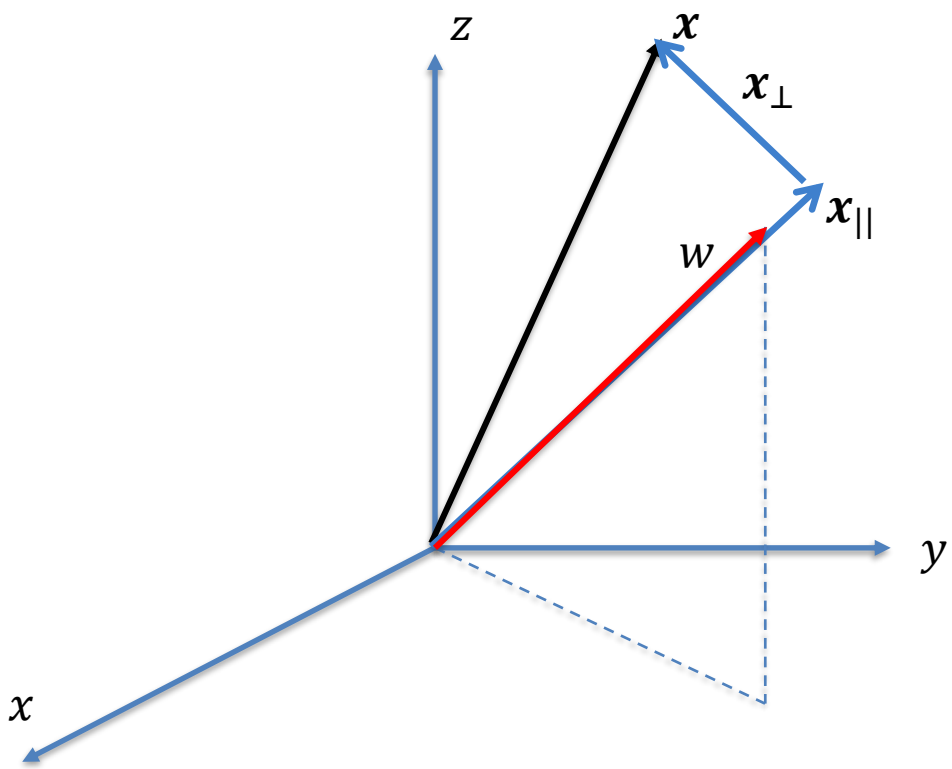
$$\boldsymbol{x} = \widehat{\boldsymbol{w}} x_{||} + \widehat{\boldsymbol{w}}_\perp x_\perp$$

$$\widehat{\boldsymbol{w}} \cdot \boldsymbol{x} = x_{||}$$

Position of $\boldsymbol{x}$ after projection
Onto weight vector $\boldsymbol{w}$

Same concept holds with N-dim vectors

Second Meaning of Linear: Linear Function Along a Line



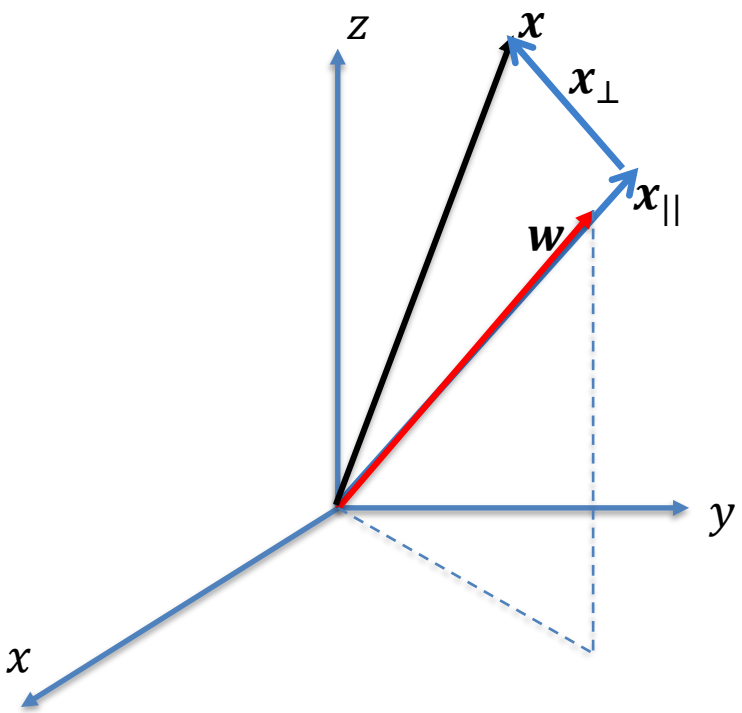
Slope

Bias/Offset

$$\boldsymbol{w} \cdot \boldsymbol{x} + b = \|w\|_2 x_{||} + b$$

Position of $\boldsymbol{x}$
After projecting onto $\boldsymbol{w}$

Second Meaning of Linear: Linear Function Along a Line

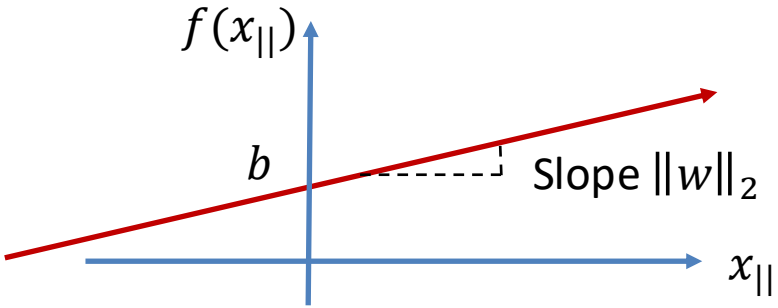


Slope

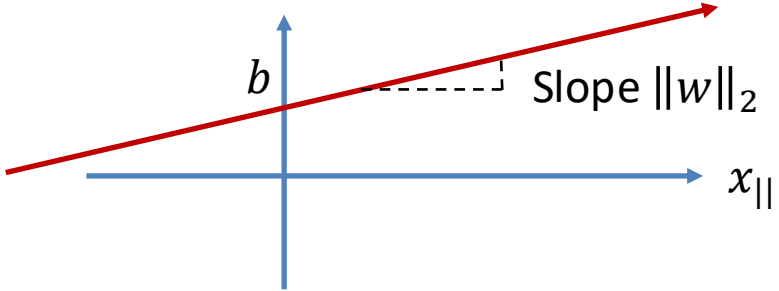
Bias/Offset

$$\boldsymbol{w} \cdot \boldsymbol{x} + b = \|w\|_2 x_{||} + b$$

Position of $\boldsymbol{x}$
After projecting onto $\boldsymbol{w}$



Second Meaning of Linear: Linear Function Along a Line



Slope

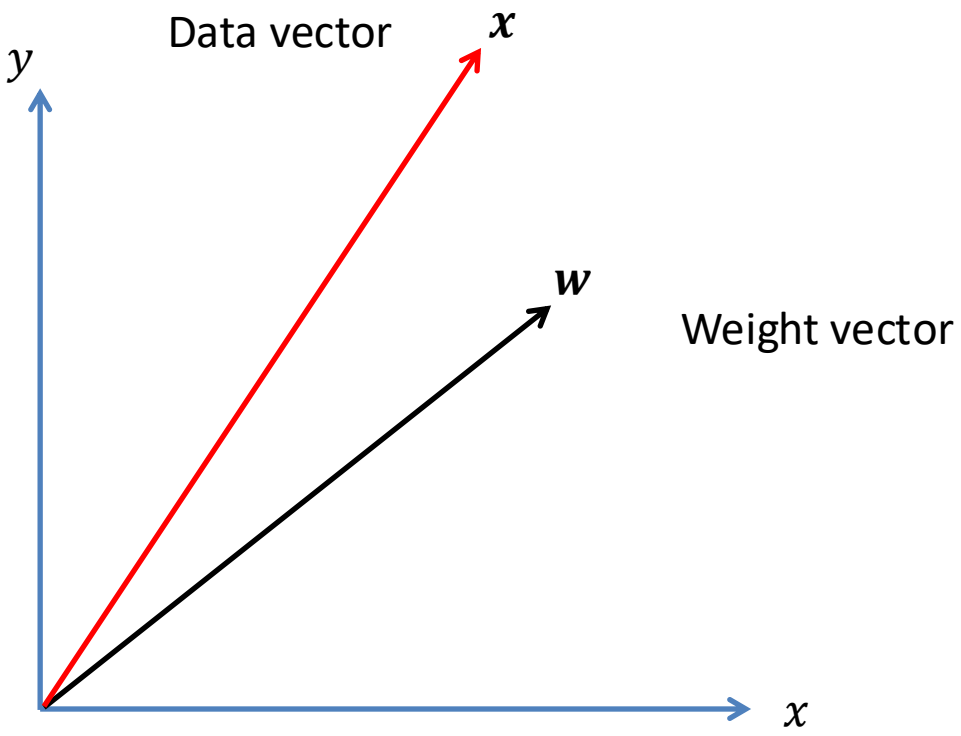
Bias/Offset

$$\boldsymbol{w} \cdot \boldsymbol{x} + b = \|w\|_2 x_{||} + b$$

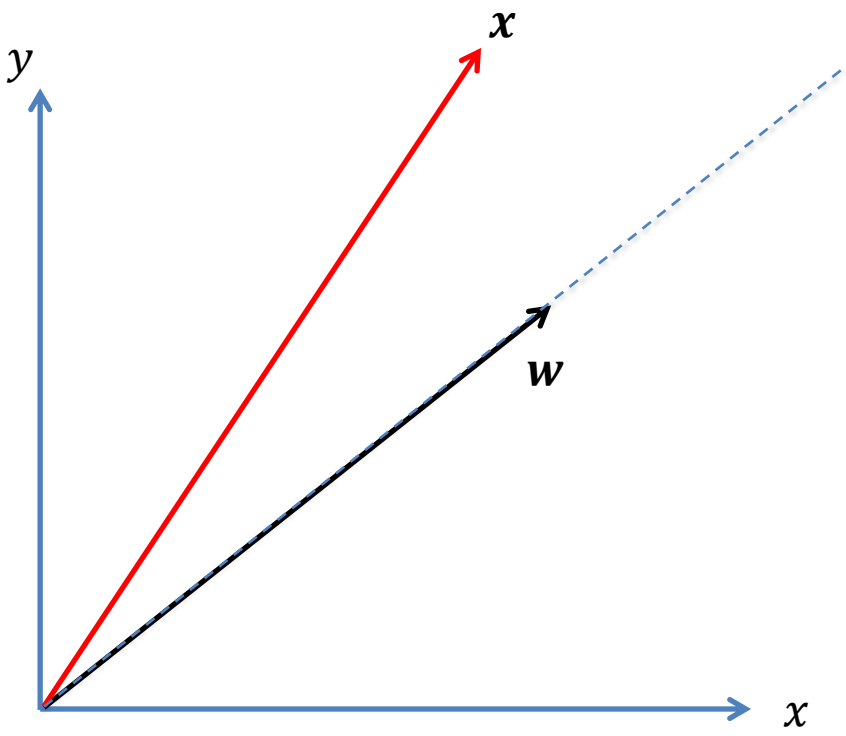
Position of $\boldsymbol{x}$
After projecting onto $\boldsymbol{w}$

- $\hat{w}$ defines direction of line onto which data x are projected
- $\|w\|_2$ defines the slope of linear model

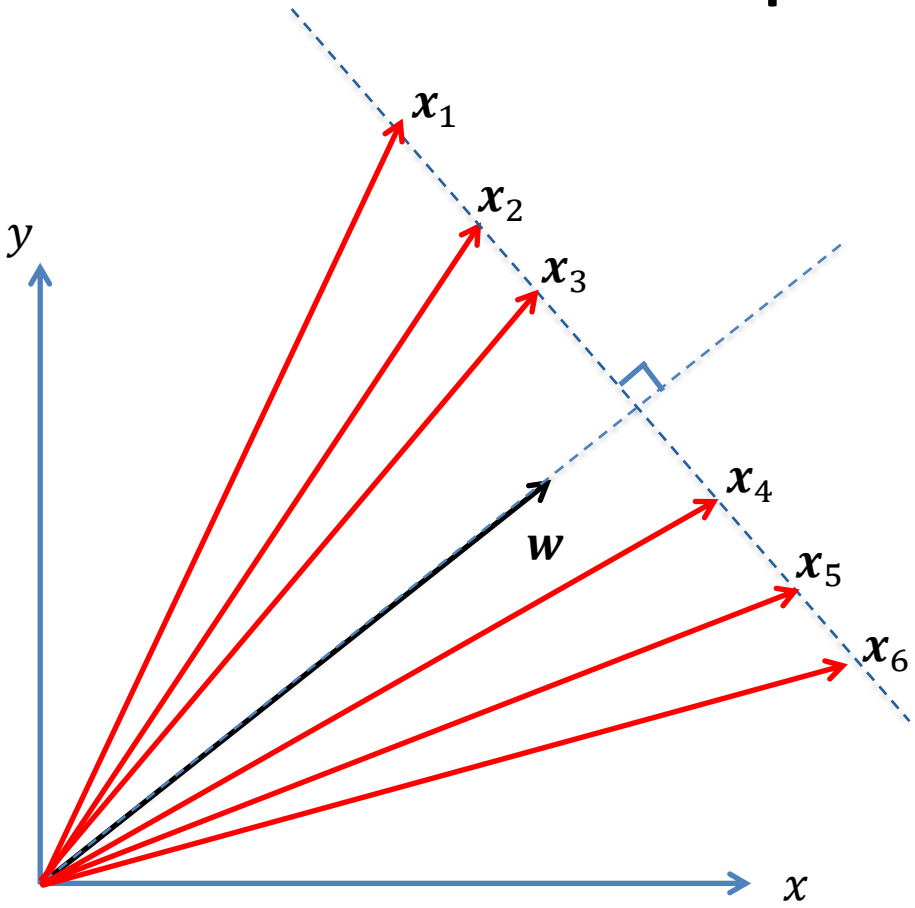
Third Perspective on Linear



Third Perspective on Linear



Third Perspective on Linear



All $\boldsymbol{x}$ on a line perpendicular to $\boldsymbol{w}$ yield the same functional Value for

$$\boldsymbol{x} \cdot \boldsymbol{w} + b$$

On Logistic Regression

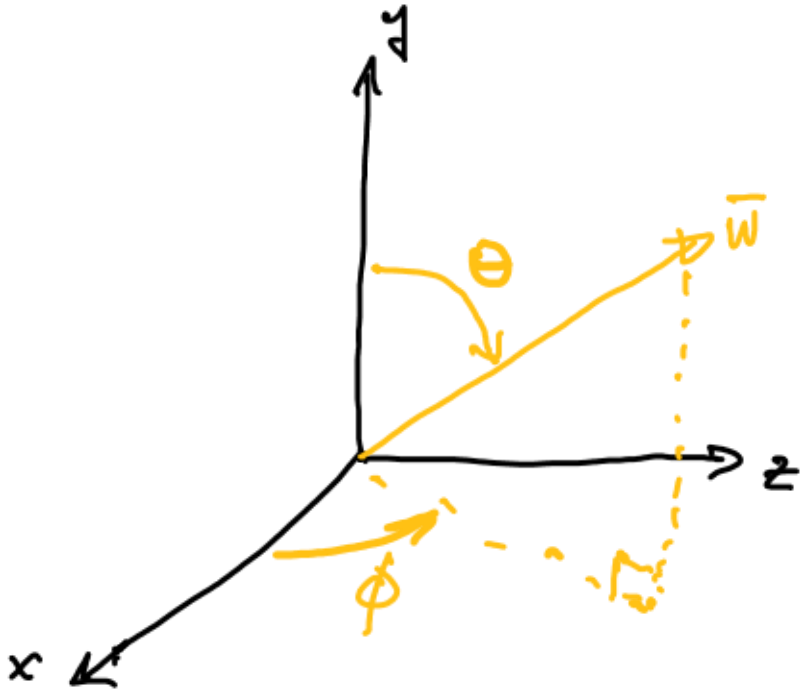
- Recall that a logistic regression model may be expressed

$$p(y = 1|x) = \sigma(w \cdot x + b)$$

- This model is linear in two of the senses we discussed:
  - The data x is projected onto a line defined by w
  - The function $\sigma(w \cdot x + b)$ is constant for x on hyperplanes perpendicular to w
- The pointwise nonlinearity $\sigma(z) = 1/[\exp(-z) + 1]$ doesn't change these aspects of what we mean by linear: Termed a *generalized* linear model (GLM)

What Does Linear Regression Assume About The Data?

- Data reside along a line in N-dim space, and a (generalized) linear model is appropriate for data projected onto this line



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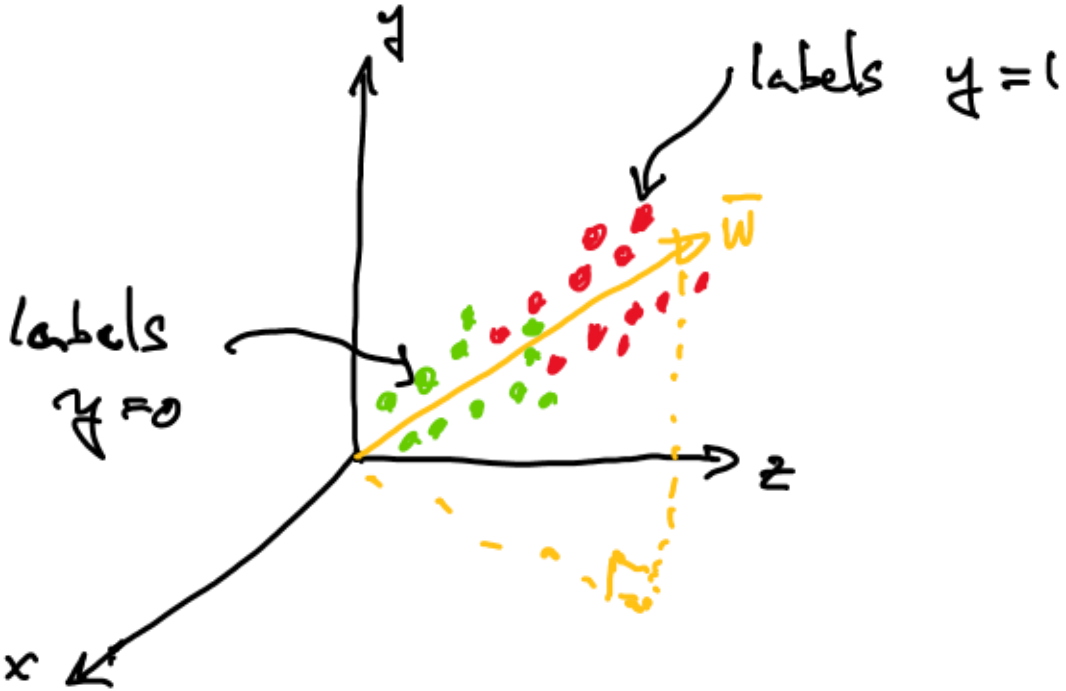

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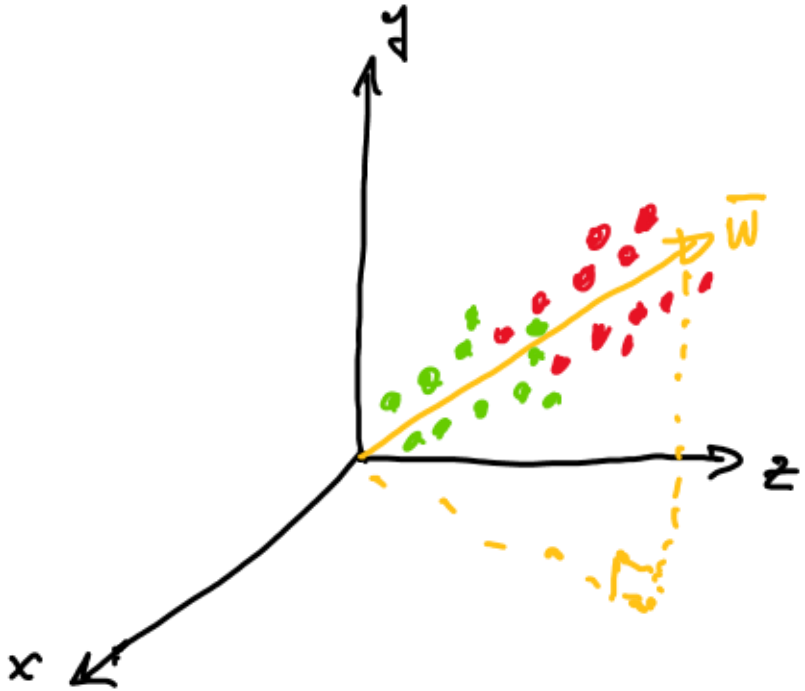
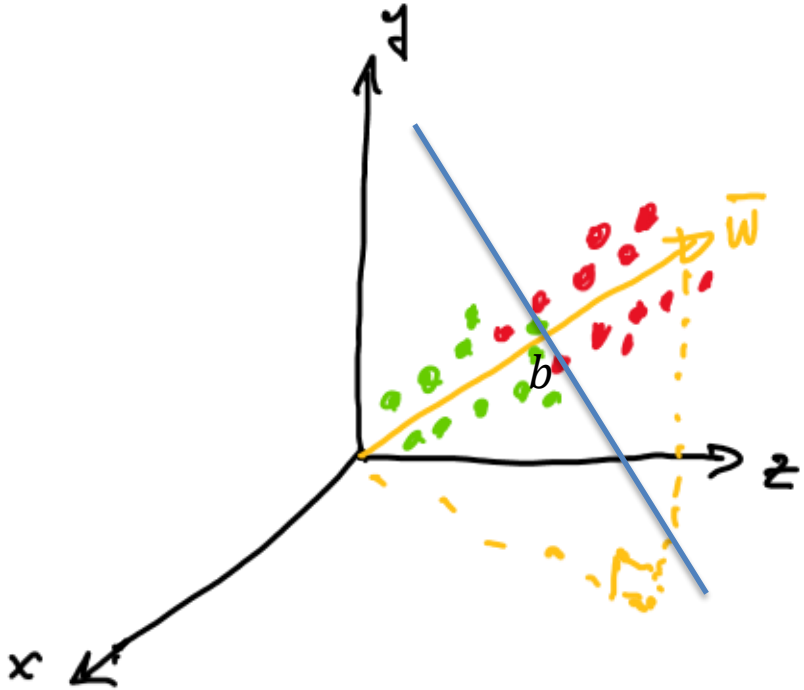


- Data is situated along a line
- The generalized linear model separates the data

What Does Linear Regression Assume About The Data?

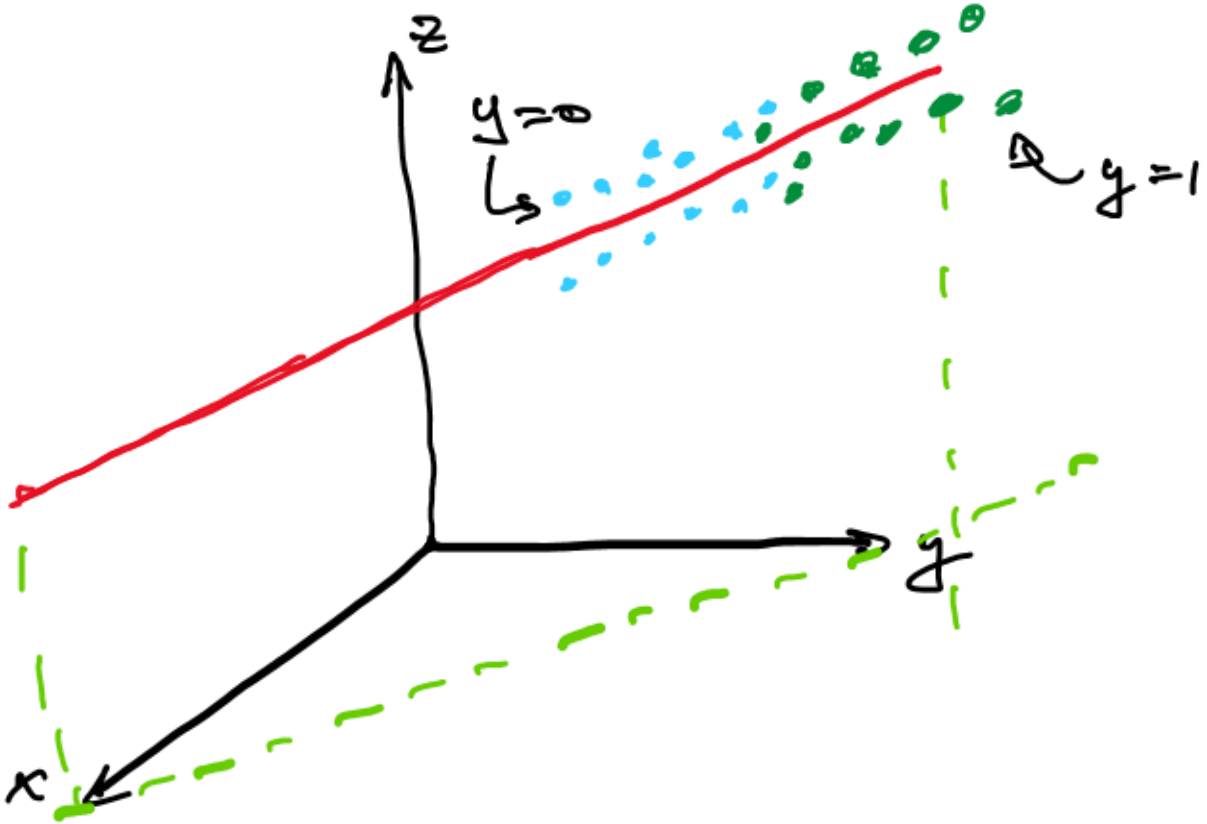
- Data reside along a line in N-dim space, and a (generalized) linear model is appropriate for data projected onto this line



Learning objective:

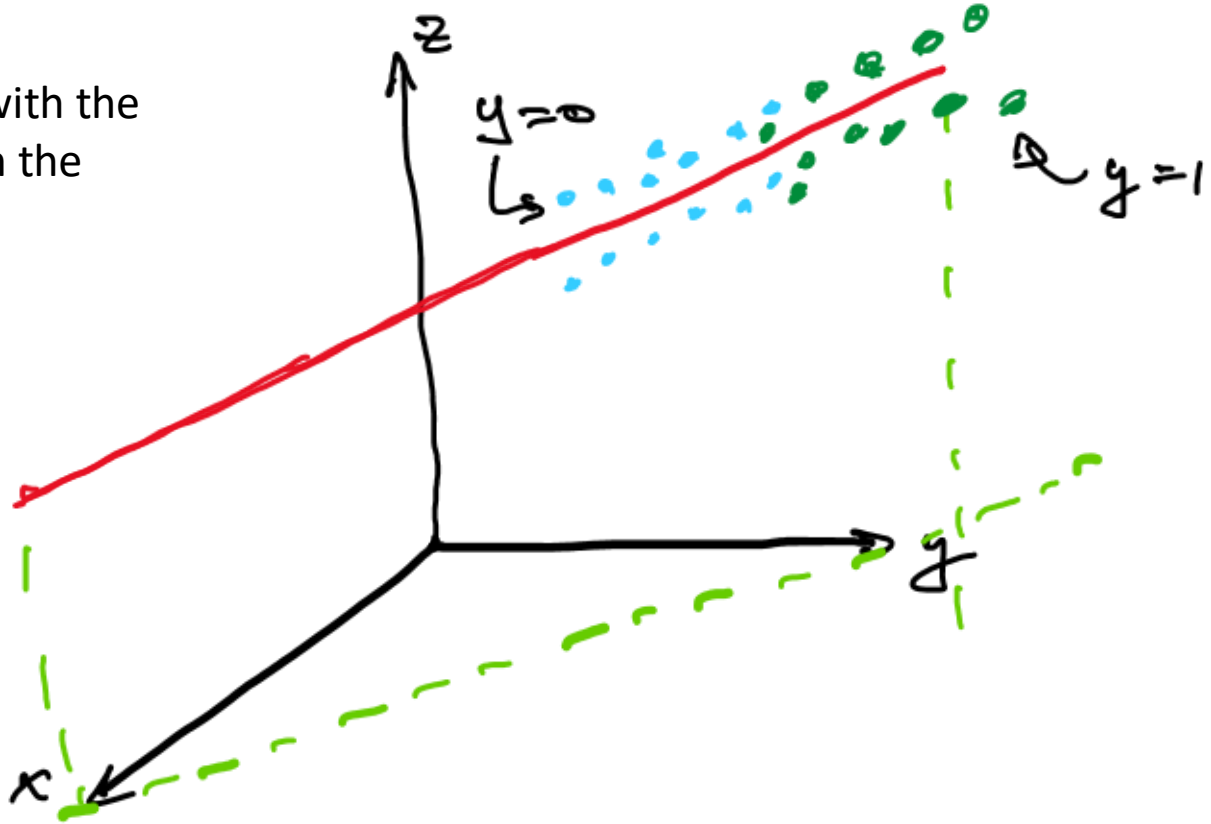
- Find the direction of $\boldsymbol{w}$, *i.e.*, $\hat{\boldsymbol{w}}$
- Find the magnitude of $\boldsymbol{w}$, *i.e.*, $\|\boldsymbol{w}\|_2$ (slope of the linear model)
- Find the bias, *i.e.*, point along the $\hat{\boldsymbol{w}}$ that *better* separates the data

What If Data Situated Along an Offset Line?



What If Data Situated Along an Offset Line?

Data is aligned with the line but not with the origin!

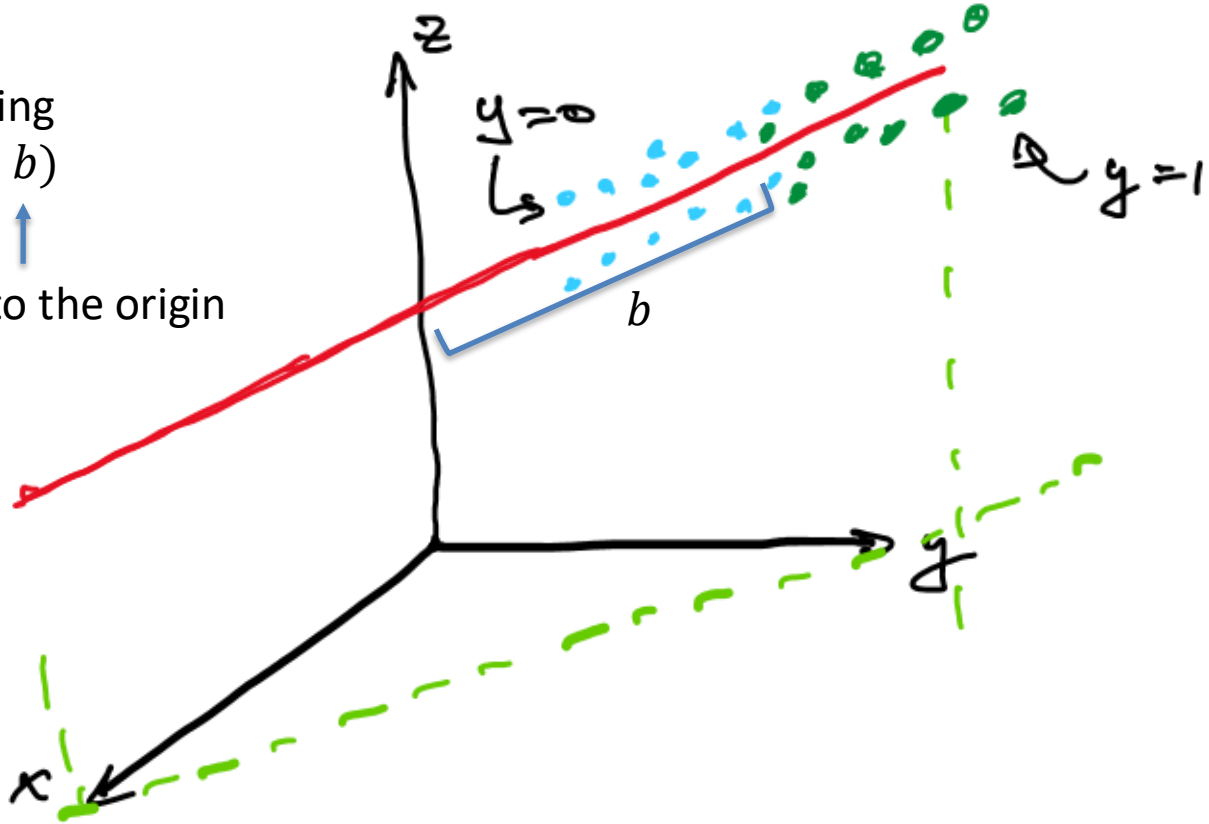


What If Data Situated Along an Offset Line?

We model this using

$$\sigma(\boldsymbol{w} \cdot \boldsymbol{x} + b)$$

Moves it to the origin



Normalize the Data!

- Assume that the labeled data are $\{(\boldsymbol{x}_i, y_i)\}_{i=1,N}$
- Calculate mean of the x_i and the standard deviation of each component of $\boldsymbol{x}_i$

Empirical mean

$$\mu = \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{x}_i$$

Coordinate j of $\boldsymbol{x}_i$

Variance of coordinate j

$$\sigma_j^2 = \frac{1}{N}\sum_{i=1}^{N} \left(x_{ij} - \mu_j\right)^2$$

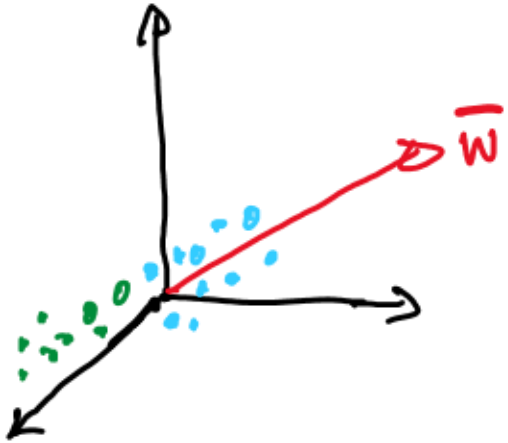
Coordinate j of $\boldsymbol{\mu}$

Normalize the Data!

- After computing data mean and component variances, normalize the data:

$$x'_{ij} = \frac{x_{ij-\mu_j}}{\sigma_{ij}}$$

- Normalized data centered about origin, situated on the line *through the origin*



Importance of Normalization: From Linear Regression to Neural Networks

- Anytime linear regression or generalized linear regression (*e.g.*, logistic regression) is performed, critical that data are normalized, to align with modeling assumptions

- Key: Data projected onto a line through origin

- In a deep neural network, generalized linear elements appear repeatedly

Projection vector

$$f(\boldsymbol{w} \cdot \boldsymbol{z} + b)$$

Nonlinear function

Features within a NN

- Essential to normalize the NN features $\boldsymbol{z}$ at the input to each layer in network

Why the Logistic Link Function?

- Fundamental to logistic regression is the logistic or sigmoid link function

$$z_i = \boldsymbol{w}^T \boldsymbol{x}_i + b$$

$$p(Y_i = 1 | X_i = x_i) = \frac{\exp(z_i)}{1 + \exp(z_i)} = \sigma(z_i)$$

- Why the sigmoidal link function, $\sigma(z)$?

Why the Logistic Link Function?

- Assume that $p(Y, X)$ is the true underlying distribution of the data, where Y is a binary random variable, $Y \in \{0,1\}$ and X is the *d*-dimensional data vector $X \in \mathbb{R}^d$

- (X, Y) are joint random variables, the distribution of which we seek to model

- From the laws of probability

$$p(Y|X) = \frac{p(Y, X)}{p(X)} = \frac{p(Y, X)}{p(Y = 0, X) + p(Y = 1, X)}$$

Why the Logistic Link Function?

$$p(Y=1|X) = \frac{p(Y=1,X)}{p(X)} = \frac{p(Y=1,X)}{p(Y=0,X)+p(Y=1,X)}$$

- We therefore have

$$p(Y=1|X) = \frac{p(Y=1,X)/p(Y=0,X)}{1+p(Y=1,X)/p(Y=0,X)}$$

Likelihood Ratio

$$p(Y=1|X) = \frac{p(Y=1,X)/p(Y=0,X)}{1+p(Y=1,X)/p(Y=0,X)}$$

- The likelihood ratio is defined as

$$L(X) = \frac{p(Y=1,X)}{p(Y=0,X)} = \frac{p(Y=1|X)p(X)}{p(Y=0|X)p(X)} = \frac{p(Y=1|X)}{p(Y=0|X)}$$

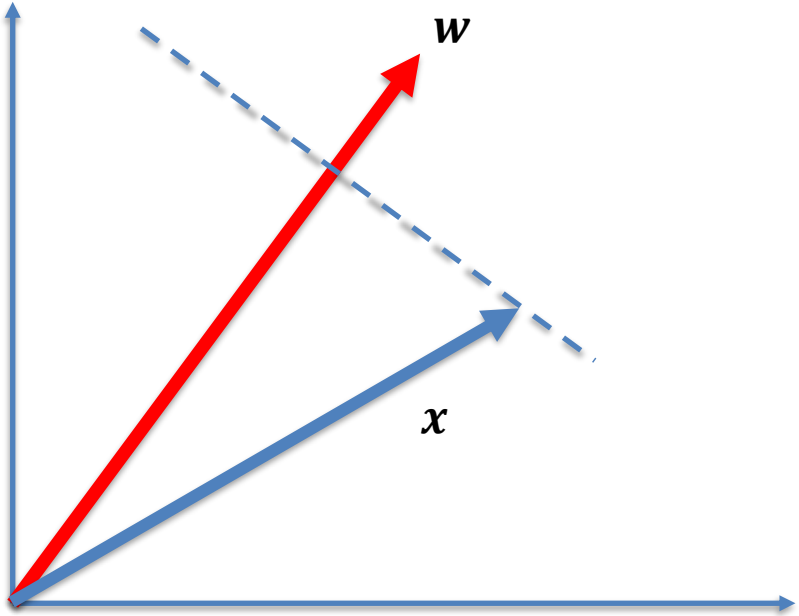
- Note that $L(X)$ is always positive, and it represents the relative likelihood of data X coming from class $Y=1$ vs class $Y=0$

Modeling the Likelihood Ratio

$$p(Y = 1|X) = \frac{L(X)}{1 + L(X)} \qquad p(Y = 0|X) = 1 - p(Y = 1|X)$$

$$L(X) = \frac{p(Y = 1|X)}{p(Y = 0|X)}$$

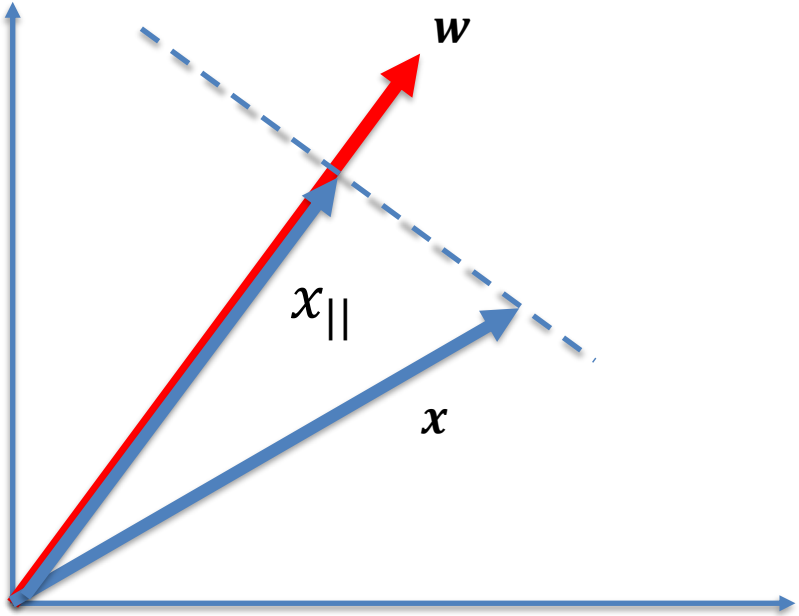
- Much of machine learning concerns modeling the likelihood ratio $L(X)$
- Simple logistic regression is based on the model $L(X) \approx \exp[w^T X + b]$
- This is a simple linear model: Linear decision surfaces perpendicular to vector $\boldsymbol{w}$ in $\mathbb{R}^d$



All $\boldsymbol{x}$ on a hyperplane perpendicular to w yield the same predicted probability for $p(Y = 1|X = \boldsymbol{x})$

$$p(Y=1|X=\boldsymbol{x}) = \frac{L(X=\boldsymbol{x})}{1+L(X=\boldsymbol{x})}$$

$$L(X=\boldsymbol{x}) \approx \exp[\boldsymbol{w}^T\boldsymbol{x}+b]$$

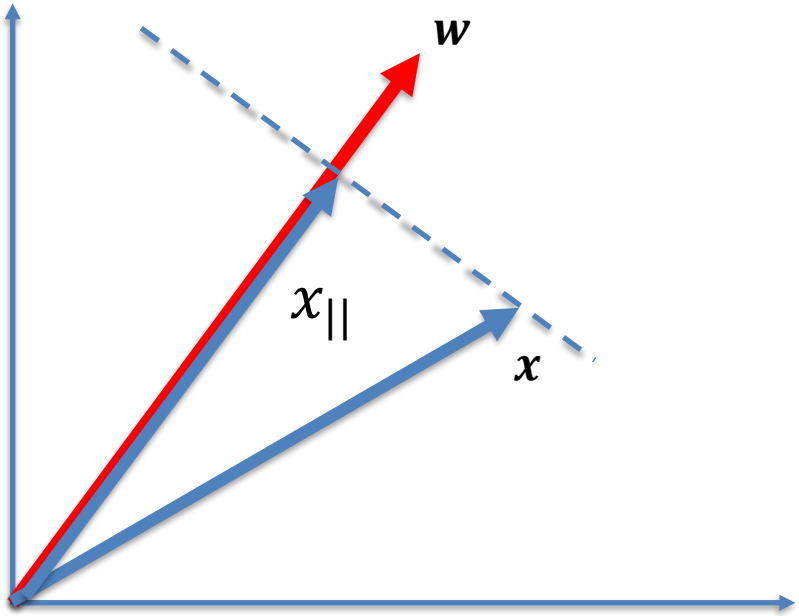


- Along the line specified by $\boldsymbol{w}$ the model of $L(X)$ is a generalized linear model

$$L(\boldsymbol{x}) = \exp[\|\boldsymbol{w}\|_2 x_{||} + b]$$

- Slope $\|w\|_2$ and intercept b

Template Matching



- Vector $\boldsymbol{w}$ defines a template for class $Y = 1$, and we perform inner-product-based matching to specify the probability of that class

$$L(x) = \exp[\|\boldsymbol{w}\|_2 x_{||} + b]$$
$$= \exp[\|\boldsymbol{w}\|_2 \widehat{\boldsymbol{w}} \cdot \boldsymbol{x} + b]$$

- Assumption of linear model in space of data X very limiting

Template Matching in Feature Space

$$p(Y = 1|X) = \frac{L(X)}{1 + L(X)} \qquad p(Y = 0|X) = 1 - p(Y = 1|X)$$

$$L(X) = \frac{p(Y = 1|X)}{p(Y = 0|X)}$$

- Move beyond linear model in the space of the data X
- Learn a nonlinear transformation $X \to \psi(X)$ where $\psi(X)$ is a learned <u>nonlinear</u> transformation of the data
- Will often model $\psi(X)$ with a neural network (to be explained in detail later)

Template Matching in Feature Space

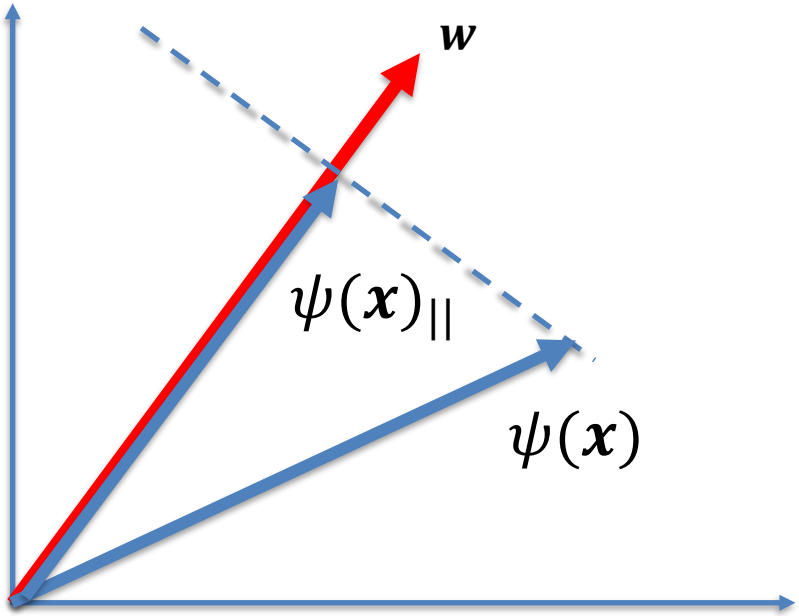
$$p(Y = 1|X) = \frac{L(X)}{1 + L(X)} \qquad p(Y = 0|X) = 1 - p(Y = 1|X)$$

$$L(X) = \frac{p(Y = 1|X)}{p(Y = 0|X)}$$

- Model likelihood ratio as $L(X = \boldsymbol{x}) = \exp[\boldsymbol{w}^T \psi(\boldsymbol{x}) + b]$

- The weight vector $\boldsymbol{w}$ is in the space of features $\psi(\boldsymbol{x})$ and the dimension of these features (dimension of vector $\boldsymbol{w}$) is typically different from the dimension of $\boldsymbol{x}$

Template Matching in Feature Space

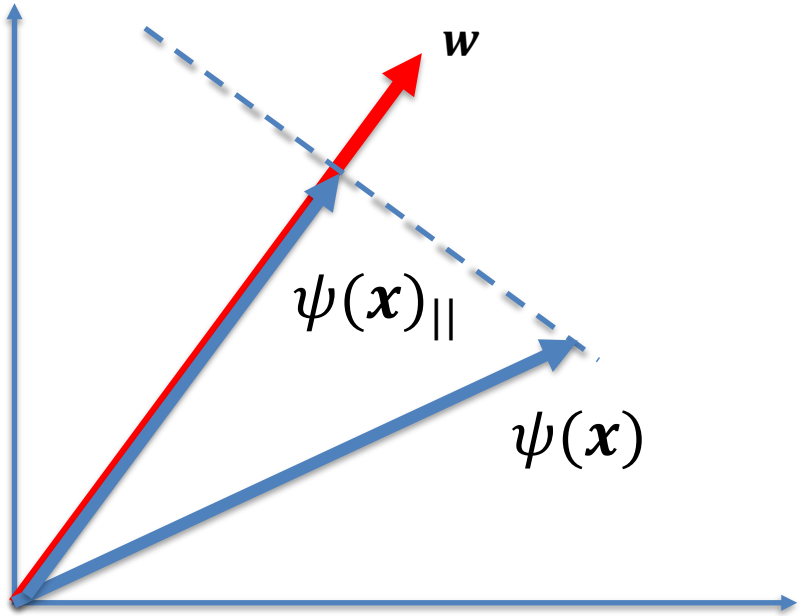


- Vector $\boldsymbol{w}$ defines a template for class $Y = 1$, and we perform inner-product-based matching in feature space $\psi(\boldsymbol{x})$

$$L(\boldsymbol{x}) = \exp[\|\boldsymbol{w}\|_2 \psi(\boldsymbol{x})_{||} + b]$$

- Because the learned function $\psi(\boldsymbol{x})$ is nonlinear, the model $\|\boldsymbol{w}\|_2\ \psi(x)_{||} + b$ is a nonlinear function of the data $\boldsymbol{x}$

The Heart of Neural Networks & Deep Learning



- We retain the idea of generalized linear models

$$L(\boldsymbol{x}) = \exp[\|\boldsymbol{w}\|_2 \psi(\boldsymbol{x})_{||} + b]$$

- However, rather than building the linear model in the data space $\boldsymbol{x}$ we build the linear model in a learned feature space $\psi(\boldsymbol{x})$

- Multi-layered ("deep") neural networks will be used to model $\psi(\boldsymbol{x})$